



PSAC 2026

Mathematics

Specimen Paper 3 of 4 | Prepared by Noukclass.com | Topics: Ratio • Data Handling • Angles • 2D Shapes

Time allowed: 1 hour 45 minutes Total Marks: 99

Centre Number: _____

Index Number: _____

Candidate's Name: _____

INSTRUCTIONS TO CANDIDATES

1. This paper contains **45** questions in three sections.
2. Write your Index Number and Name in the spaces provided above.
3. Answer **all** questions.
4. Write your answers in the spaces provided in this booklet.
5. Show **all** working clearly. Marks may be awarded for working even if the final answer is wrong.
6. Do **not** use red, green or black ink.
7. Diagrams are **not** drawn to scale unless stated otherwise.
8. Calculators are **not** allowed.

Section	Questions	Marks Available	Marks Awarded	Examiner
A	1–18	36		
B	19–28	20		
C	29–45	43		
Total		99		

SECTION A — Short Answer Questions

Questions 1–18 | 36 marks | Write your answers clearly in the spaces provided.

1. Write the ratio **24 : 36** in its simplest form.

2 marks

Answer: _____

2. Share **Rs 560** in the ratio **3 : 5**.

2 marks

Smaller share: Rs _____ Larger share: Rs _____

3. In a class, the ratio of boys to girls is **3 : 4**. There are **12 boys**. How many girls are there?

2 marks

Answer: _____ girls

4. What is the name of an angle that is exactly **90°**? What about one that is between 90° and 180°?

2 marks

90° angle: _____ **between 90° and 180°:** _____

5. Find the missing angle. The angles in a triangle are **65°**, **48°** and **x°**.

2 marks

$x = \underline{\hspace{2cm}}^\circ$

6. Angles on a straight line add up to **180°** . Find angle y if the other angle on the line is **123°** . 2 marks

$y = \underline{\hspace{2cm}}^\circ$

7. A quadrilateral has angles of **110°** , **85°** , **95°** and z° . Find z . 2 marks

$z = \underline{\hspace{2cm}}^\circ$

8. How many lines of symmetry does a regular hexagon have? 2 marks

Answer:

How many lines of symmetry does an equilateral triangle have?

Answer:

9. Name the 2D shape that has **5 sides** and **5 angles**. What is the sum of its interior angles? 2 marks

Shape: **Sum of angles:** $^\circ$

10. Look at this frequency table: 2 marks

Colour	Frequency
Red	8
Blue	12
Green	5
Yellow	9

What is the total number of items? What is the most popular colour?

Total: **Most popular:**

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11. Simplify the ratio **1.2 : 0.8**. Then write it in the form $n : 1$.

2 marks

Simplified: _____ **Form $n : 1$:** _____

12. A rectangle has all four angles equal to **90°** . What is the special name for a rectangle where all sides are equal?

2 marks

Answer: _____

How many pairs of parallel sides does a trapezium have?

Answer: _____

13. This pictogram shows books read by children in one month. Each  represents 4 books.

2 marks

Child	Books read
Asha	  
Ben	    
Celia	 

How many books did Ben read? What is the total for all three children?

Ben: _____ **Total:** _____

14. Find the mean (average) of these numbers: **14, 18, 22, 10, 16**

2 marks

Mean = _____

15. A recipe uses flour and sugar in the ratio **5 : 2**. If you use **350 g of flour**, how much sugar do you need?

2 marks

Sugar = _____ g

16. What is the size of each interior angle of a regular polygon with **4 sides** (a square)? What about a regular polygon with **6 sides**? 2 marks

Square interior angle: _____ ° Regular hexagon interior angle: _____ °

17. The range of a data set is **18**. The smallest value is **7**. What is the largest value? Find the median of: **3, 7, 9, 12, 15**. 2 marks

Largest value: _____ Median: _____

18. How many **degrees** are in a full turn? How many degrees are in three-quarters of a turn? 2 marks

Full turn: _____ ° Three-quarter turn: _____ °

SECTION B — Multiple Choice Questions

Questions 19–28 | 20 marks | Tick (✓) ONE box for each question. Each question is worth 2 marks.

Questions 19–20

19. Which ratio is equivalent to 3 : 5?

- ☐ A 6 : 15
- ☐ B 9 : 15
- ☐ C 12 : 25
- ☐ D 15 : 20

20. The sum of angles in a triangle is always:

- ☐ A 90°
- ☐ B 270°
- ☐ C 180°
- ☐ D 360°

Questions 21–22

21. The mean of 5, 10, 15, 20 is:

- ☐ A 10
- ☐ B 12.5
- ☐ C 13
- ☐ D 15

22. How many lines of symmetry does a rectangle have?

- ☐ A 1
- ☐ B 2
- ☐ C 4
- ☐ D 0

Questions 23–24

23. Two angles of a triangle are 55° and 75° . The third angle is:

- ☐ A 40°
- ☐ B 50°
- ☐ C 60°
- ☐ D 70°

24. Rs 900 is shared in ratio 2 : 7. The smaller share is:

- ☐ A Rs 100
- ☐ B Rs 180
- ☐ C Rs 200
- ☐ D Rs 250

Questions 25–26

25. A regular pentagon has interior angles each measuring:

- ☐ A 108°
- ☐ B 120°
- ☐ C 90°
- ☐

26. A bar chart shows scores: Mon=20, Tue=35, Wed=25, Thu=40, Fri=30. The modal (most frequent) day score is:

- ☐ A Monday
- ☐ B Tuesday
- ☐

☐ **D** 72°

☐ **C** Thursday

☐ **D** Friday

Questions 27–28

27. An angle of 210° is called:

- ☐ **A** Acute
- ☐ **B** Obtuse
- ☐ **C** Reflex
- ☐ **D** Right

28. The median of 3, 6, 9, 12, 15, 18, 21 is:

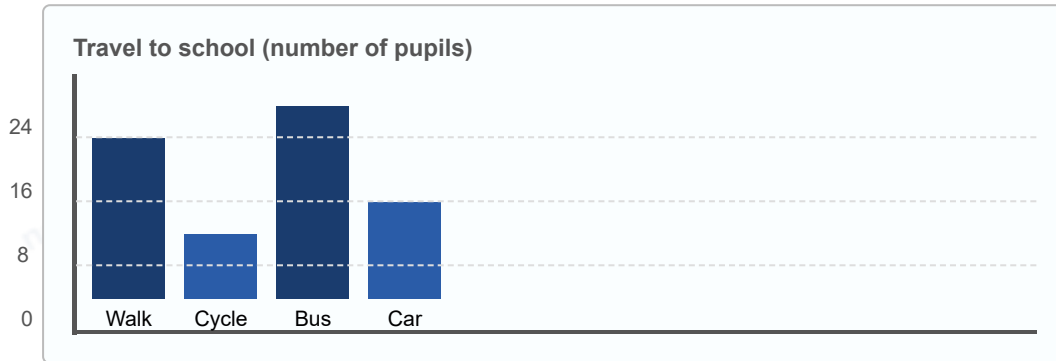
- ☐ **A** 9
- ☐ **B** 12
- ☐ **C** 13.5
- ☐ **D** 15

SECTION C — Structured Questions

Questions 29–45 | 43 marks | Show ALL working. Marks are awarded for method as well as the answer.

29. The bar chart below shows the number of pupils who travel to school by different methods.

3 marks



- (a) How many more pupils travel by bus than by cycle?

Answer: _____ more pupils

- (b) What is the total number of pupils?

Answer: _____ pupils

- (c) What fraction travel by car? Give your answer in simplest form.

Fraction = _____

30. Orange squash is made by mixing **1 part concentrate** to **4 parts water**.

3 marks

(a) Write the ratio of concentrate to water in its simplest form.

Ratio: _____

(b) How much concentrate is needed to make **750 ml** of squash?

Concentrate = _____ ml

(c) How much water is needed?

Water = _____ ml

31. The scores of 6 students in a test are: **72, 58, 85, 63, 79, 91**.

2 marks

(a) Find the range.

Range = _____

(b) Find the mean score.

Mean = _____

32. Look at triangle PQR below. All angles are given in terms of x .

3 marks

Angle P = $2x$ Angle Q = $x + 20^\circ$ Angle R = $x - 5^\circ$

(a) Write an equation using the fact that angles in a triangle sum to 180° .

(b) Solve to find x .



$x =$ _____ $^\circ$

(c) Find all three angles.

$P =$ _____ $^\circ$ $Q =$ _____ $^\circ$ $R =$ _____ $^\circ$

33. A pictogram shows trees planted each year. Each  represents **6 trees**.

2 marks

Year	Trees planted
2022	  
2023	    
2024	   

- (a) How many trees were planted in 2023? (b) How many trees were planted in total?

(a) _____ (b) _____

34. The table shows daily rainfall in mm for one week:

3 marks

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Rainfall (mm)	4	0	12	8	5	16	7

- (a) On which day did it rain most?

Answer: _____

- (b) Find the total rainfall for the week.

Total = _____ mm

- (c) Find the mean daily rainfall.

Mean = _____ mm

35. Concrete is made by mixing cement, sand and gravel in the ratio **1 : 2 : 4**. If **7 kg of cement** are used:

2 marks

(a) How much sand is needed?

Sand = _____ kg

(b) What is the total mass of concrete?

Total = _____ kg

36. Draw **all** lines of symmetry on this isosceles triangle (describe them if you cannot draw). How many does it have?

2 marks



Number of lines of symmetry: _____

37. Here are five numbers: **14, 9, 22, 17, 8, 22, 11**

3 marks

(a) Find the mode.

Mode = _____

(b) Find the median.

Median = _____

(c) Find the mean.

Mean = _____

38. A map is drawn at a scale of **1 : 50,000**. On the map, two towns are **4 cm** apart.

3 marks

(a) What is the real distance in cm?

Real distance = _____ cm

(b) Convert this to kilometres.

Real distance = _____ km

(c) A lake is 3 km long in reality. What length should it be drawn on the map?

Map length = _____ cm

39. Two angles of a quadrilateral are 100° and 80° . The other two angles are equal. Find each of these equal angles.

2 marks

Each equal angle = _____ $^\circ$

40. The ages of pupils in a class are recorded. The frequency table is:

3 marks

Age	Frequency	Age $\times$ Freq.
10	6	
11	14	
12	10	
Total		

(a) Complete the table.

(b) Find the mean age.

Mean age = _____

(c) What is the modal age?

Modal age = _____

41. Two boys share sweets in the ratio $5 : 3$. The boy with more gets **35 sweets**.

3 marks

(a) How many sweets does the other boy get?

Answer: _____ sweets

(b) How many sweets are there in total?

Total = _____ sweets

(c) If a third boy joins and the sweets are re-shared in the ratio 5 : 3 : 2, how many does each boy get? (Total = 56 sweets)

5 parts: _____ 3 parts: _____ 2 parts:

42. Name each type of angle and give its size:

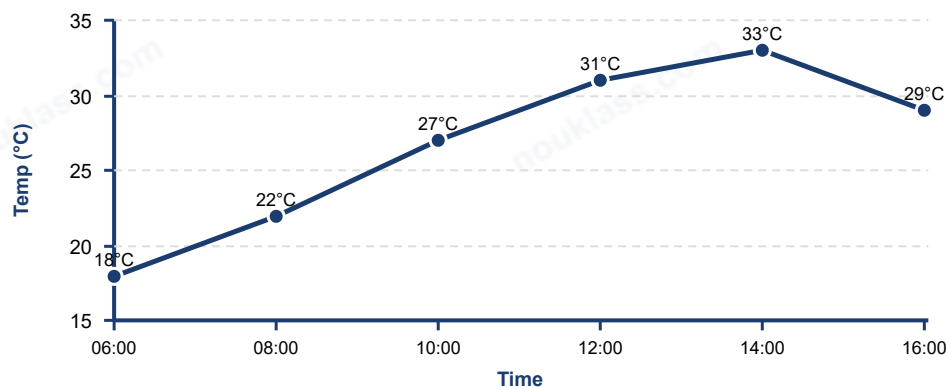
2 marks

(a) An angle that is less than 90° is called: _____

(b) An angle that is exactly 180° is called: _____

43. The line graph shows the temperature ($^\circ\text{C}$) measured every 2 hours on a Thursday:

3 marks



(a) What was the temperature at noon?

Answer: _____ $^\circ\text{C}$

(b) By how many degrees did the temperature rise from 06:00 to 14:00?

Rise = _____ $^\circ\text{C}$

(c) What is the mean temperature over the six readings?

Mean = _____ $^\circ\text{C}$

44. Describe the properties of a parallelogram. Name one special type of parallelogram.

2 marks

Special type: _____

- 45.** Three angles around a point are $2x$, $3x$ and $x + 60^\circ$. Find x and all three angles. (Angles at a point = 360°)

2 marks

x = _____ ° **Angles:** _____ ° _____ °

Mark Scheme — Noukclass.com Specimen Paper 3 of 4 — 2026

This mark scheme is provided for self-checking and teacher reference only. For method marks, show all working. Prepared by Noukclass.com based on MES syllabus and past paper analysis. Not an official MES document.

Q	Answer	Marks	Notes
1	2 : 3	2	HCF of 24 and 36 = 12; $24/12:36/12 = 2:3$
2	Rs 210 : Rs 350	2	Total parts=8; unit=70; $3 \times 70=210$; $5 \times 70=350$
3	16 girls	2	$12 \div 3=4$ (1 part); $4 \times 4=16$
4	Right angle; Obtuse angle	2	Standard angle names
5	67°	2	$180-65-48=67$
6	57°	2	$180-123=57$
7	70°	2	$360-110-85-95=70$
8	Regular hexagon: 6 lines; Equilateral triangle: 3 lines	2	Standard symmetry facts
9	Pentagon; 540°	2	Sum = $(5-2) \times 180 = 540$
10	Total: 34; Most popular: Blue (12)	2	$8+12+5+9=34$
11	3 : 2 (simplest); 1.5 : 1	2	$1.2:0.8 = 12:8 = 3:2$; divide by 2 for n:1
12	Square; 1 pair	2	A square is a rectangle with equal sides; trapezium has 1 pair parallel sides
13	Ben: 20 books; Total: 40 books	2	$5 \text{ symbols} \times 4 = 20$; $(3+5+2) \times 4=40$
14	16	2	$(14+18+22+10+16) \div 5=80 \div 5=16$
15	140 g	2	$350 \div 5=70$ (1 part); $70 \times 2=140$
16	Square: 90° ; Regular hexagon: 120°	2	$(4-2) \times 180 \div 4=90$; $(6-2) \times 180 \div 6=120$
17	Largest = 25; Median = 9	2	$7+18=25$; middle value of 3,7,9,12,15 is 9
18	360° ; 270°	2	Full turn= 360 ; $3/4 \times 360=270$
19	B — 9 : 15	2	9:15 simplifies to 3:5 (divide by 3)
20	C — 180°	2	Angle sum of triangle = 180°

21	B — 12.5	2	$(5+10+15+20)\div 4=50\div 4=12.5$
22	B — 2	2	Rectangle has 2 lines of symmetry (horizontal and vertical)
23	B — 50°	2	$180-55-75=50$
24	C — Rs 200	2	$900\div 9=100$ per part; $2\times 100=200$
25	A — 108°	2	$(5-2)\times 180\div 5=540\div 5=108$
26	C — Thursday (score 40)	2	Highest bar = Thursday
27	C — Reflex	2	Reflex: between 180° and 360°
28	B — 12	2	7 values; middle (4th) value = 12
29	(a) 16 more (b) 64 pupils (c) $12/64 = 3/16$	3	Bus=24, Cycle=8, $24-8=16$; Walk=20, Cycle=8, Bus=24, Car=12; $12/64=3/16$
30	(a) 1:4 (b) 150 ml (c) 600 ml	3	$1+4=5$ parts; $750\div 5=150$; $150\times 4=600$
31	(a) Range=33 (b) Mean=74.67	2	$91-58=33$; $(72+58+85+63+79+91)\div 6=448\div 6\approx 74.7$
32	(a) $2x+x+20+x-5=180 \rightarrow 4x+15=180$ (b) $x=41.25^\circ$ (c) $P=82.5^\circ$, $Q=61.25^\circ$, $R=36.25^\circ$	3	$4x=165$; $x=41.25$
33	(a) 30 trees (b) 72 trees	2	$5\times 6=30$; $(3+5+4)\times 6=72$
34	(a) Saturday (b) 52 mm (c) $52\div 7\approx 7.4$ mm	3	$4+0+12+8+5+16+7=52$
35	(a) 14 kg (b) 49 kg	2	1 part=7; sand= $2\times 7=14$; total= $7(1+2+4)=49$
36	1 line of symmetry	2	Isosceles triangle: 1 line through apex to midpoint of base
37	(a) 22 (b) 14 (c) 14.71 ($103\div 7$)	3	Ordered: 8,9,11,14,17,22,22; mode=22; median=14; mean= $103/7$
38	(a) 200,000 cm (b) 2 km (c) 6 cm	3	$4\times 50000=200000\text{cm}=2\text{km}$; $3\text{km}=300000\text{cm}$; $300000\div 50000=6$
39	90°	2	$360-100-80=180$; $180\div 2=90$
40	(a) 60,154,120; Total=30, total $\times$ freq=334 (b) Mean= $334\div 30\approx 11.1$ (c) Modal age=11	3	Highest frequency is 14 (age 11)
41	(a) 21 (b) 56 (c) 28, 16.8, 11.2	3	$35\div 5=7$; $3\times 7=21$; total=56; $56\div 10=5.6$ per part
42	(a) Acute (b) Straight angle	2	Standard angle names

43	(a) 31°C (b) 15°C (c) 26.7°C	3	$33-18=15$; $(18+22+27+31+33+29)\div 6=160\div 6\approx 26.7$
44	Opposite sides equal and parallel; angles not necessarily 90°. Special types: rectangle, rhombus, square.	2	Accept any one correct special type
45	$x=50^\circ$; Angles: 100°, 150°, 110°	2	$2x+3x+x+60=360$; $6x=300$; $x=50$